

Capstone Assignment 1

*** I put the assignments for the next meeting in boldface ***

For our next meeting

- Find out what you have found on the web - interesting resources
- Discuss the requirements document
- Discuss the technology

Tools

Github version control

- **get an account on github.com and create a repository the load and save files in the repository** there should be some simple youtube tutorials on how to use github. Don't do anything too complicated just get familiar with it

Let me know your github username/email

- I will get you an account log into **AI Opebwebui server and Jupyter compute server. Just poke around, and tell me your impressions**

Get bash and ssh working on your MacOS

If you run into any problems next week, we can share screens and I can talk you through any trouble you may have

Requirements

Think about what will go into the requirements doc

Before we start coding we will need a **requirements doc** that says * What the product is? * Who will use it? * What tools we will use

I think we would want to finalize the doc by meeting 3, but for meeting 2, we will just discuss some ideas. So I would like you to

come up with some ideas for our next meeting

You can come up with several different ideas for us to discuss. or if you are sure you want to do something, you can focus on that and start thinking about how to put together the site

Also think about the type of site you would like to create

- A forum (think facebook)
- A dashboard (take a lot of data and put it together)
- A tool (something that would let you calculate something)

One thing that you should think about it is how to use AI? AI is very good at putting together different sources of information (like what stars are visible from SG assuming this much light pollution)

Also if you want to focus on light pollution, I will ask you to do a mini-report on how light pollution is measured (since I don't know)

Technical design

Go into your favorite browser look at developer mode

surf the web for ideas and come up with ideas on the tools you would like to use

- look at web pages that are similar that what you would like to do ** Google for "citizen science" ** Look for people that are doing things you are interested in near Singapore
- The platforms that you would like to use.
- **Research programming languages, particularly python**
- **Research AI tools for generating code**
- Take a look at ****Jupyter Viola and Marimo on the compute server****
- Look at what is available in **AI Tools**
- A lot will depend on what you want to learn first. So look at the youtube tutorials and think about 1) what can you get done in 15 weeks and 2) what can you expand on

Think about what you can get done

Make sure you keep a file of all of things you are looking at

In research its okay to use someone elses ideas as long as you let people know where you got the idea from. So you want to keep a log of the sites you have seen so that you can give credit to them

Scientific publications

take a look at the websites ORCID and Zenmodo

No need to log in. Just take a look

Find out where the preprints archives are for the semiconductor work that you are doing

- Every field has there publishing hub
- In astrophysics **<http://arxiv.org/>** and **<https://adswww.harvard.edu/>**
Please visit those sites and let me know if you find anything interesting
- In High Energy Physics it is <http://inspirehep.net/>
- My entry is <https://inspirehep.net/authors/1911835>